IMO(9^{th}) – 1967

First Day

1. ABCD is a parallelogram; $AB = a, AD = 1$, α is the size of the $\angle DAB$, and the three angles of the triangle ABD are acute. Prove that the four circles K_A, K_B, K_C, K_D each of radius 1, whose centers are the vertices A, B, C and D, cover the parallelogram if and only if $a \le \cos\alpha + \sqrt{3}\sin\alpha$.
2. Exactly one side of a tetrahedron is of length greater than 1. Show that its volume is less than or equal to 1/8.
3. Let k, m and n be positive integers such that $m+k+1$ is a prime number greater than $n+1$. Write c_s for $s(s+1)$. Prove that the product $(c_{m+1}-c_k)(c_{m+2}-c_k)........(c_{m+n}-c_k)$ is divisible by the product $c_1c_2........c_n$.

Second Day

4. The triangle $A_0B_0C_0$ and $A'B'C'$ have all their angles acute. Describe how to construct one of the triangle ABC similar to $A'B'C'$ and circumscribing $A_0B_0C_0$ (so that A, B, C corresponds to corresponds to A', B', C' and AB passes through C_0, BC passes through A_0 and CA passes through B_0). Among these triangles ABC describe, and prove how to construct the triangle with maximum area.
5. Consider the sequence (C_n):
   $C_1 = a_1 + a_2 + ...+ a_8$,
   $C_2 = a_1^{\ 2} + a_2^{\ 2} + ...+ a_8^{\ 2}$,
   …………………………………
   $C_n = a_1^{\ n} + a_2^{\ n} + ...+ a_8^{\ n}$
   …………………………………
   Where $a_1, a_2, ..., a_8$ are real numbers, not all equal to zero. Given that among the numbers of the sequence (C_n) there are infinitely many equal to zero, determine all values of n for which $C_n = 0$.
6. In sports competition lasting n days there are m medals to be own. On the first day, one medal and 1/7 of the remaining m – 1 medals are own. On the second day, 2 and 1/7 of the remainder are won. And so on. On the nth day exactly n medals are won. How many days did the competition last and what was the total number of medals?

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